# Aaron Caballero Rojas

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## EDUCATION

**University of Florida**, Gainesville, FL — ***Expected May 2028***

Bachelor of Science in Aerospace and Mechanical Engineering | GPA: 3.91

**Broward College**, Coconut Creek, FL

Associate of Arts | GPA: 3.92 — ***May 2025***

## TECHNICAL SKILLS

CAD: SolidWorks, AutoCAD, Autodesk Fusion | Analysis: ANSYS Mechanical, ANSYS Fluent, XFLR5
Programming: MATLAB, Python, Excel

## WORK EXPERIENCE

**AutoCAD 2D Drafter | AED Consulting** — ***June 2025–Present***

- Produced over 200 AutoCAD drafts for aluminum roofs, pool screen enclosures, screen rooms, and floor plans, averaging about 45 minutes per assignment.
- Developed reusable AutoCAD templates and organized drawing details by beam and post configuration to streamline recurring drafting tasks.
- Coordinated assignments with senior staff and delivered project drawings to established deadlines.

## ENGINEERING LEADERSHIP

**New Member Project Chair | AeroGator, UF** — ***January–April 2026***

- Led two teams of new members through aircraft design and mission-feasibility analysis under mass, volume, payload, and endurance requirements.
- Taught lift and drag fundamentals and SolidWorks and Fusion workflows; guided teams in coordinating component dimensions and assembly interfaces.
- Guided teams in using aerodynamic estimates and mission requirements to obtain energy needs, select motors, and batteries.

## ENGINEERING PROJECTS

**Liquid Bi-propellant Rocket Engine Design | Florida Rocket Lab, UF** — ***September-October 2026***

- Built Excel calculations to size a preliminary pressure-fed LOX/IPA engine for a 1,000 lbf thrust target and 5 s burn, including chamber and nozzle geometry.
- Drafted the propellant feed-system P&ID and estimated pressure losses across fluid components.
- Generated a bell nozzle contour using the method of characteristics, improving thrust performance by 1.3% when compared with a standard Rao nozzle.
- Designed a six-element coaxial swirl injector and modeled the injector, manifolds, and distribution plates in SolidWorks
- Implemented an augmented spark igniter fueled by GOX and IPA, loaded on the central axis of the injector
- Performed preliminary chamber heat-transfer analysis and incorporated injector-face film cooling into the design.
- Used tables from Parker's O-ring Handbook to calculate O-ring and groove geometry for sealing.
- Calculated bolt and joint analysis using Excel
- Obtained supplier quotes and established a Bill of Materials for all parts of the assembly

**UAV Airframe Designer | AeroGator, UF** — ***September 2025–May 2026***

- Compared candidate airfoils in XFLR5 for a terrain-mapping UAV intended for a laboratory at the University of Ghana.
- Imported airfoil coordinates into SolidWorks, contributed to the conventional tail design and researched tapered-wing trade-offs for low-speed flight.
- Evaluated carbon fiber and fiberglass for aircraft construction during the design phase.

**Aircraft Designer and Electronics Contributor | IAM3D, UF** — ***September 2025–May 2026***

- Designed RC-aircraft fuselage geometry and a four-rotor drone base plate in Autodesk Fusion.
- Connected a FlySky radio-control receiver to an Arduino; contributed to motor and servo control integration for the aircraft.

**Solar Cell Thermal Analysis | Independent Project** — ***November 2025***

- Modeled a solar cell and 72-cell panel in SolidWorks; used the single-cell model for an exploratory ANSYS steady-state thermal analysis within available mesh limits.
- Generated ANSYS APDL input files for 10 prescribed irradiance levels using MATLAB and plotted exported temperature and heat-flux results.